

PYTHON-I T1 QUESTION PAPER GENERATION GUIDELINES

Faculty-only operational guide • LJJET Semester-I Batch 2026

Confidentiality: This document contains faculty-only blueprint and compliance controls. Do not circulate it to students.

1. Paper Identity and Schedule

Item	Requirement
Institute	L. J. Institute of Engineering and Technology, Ahmedabad
Programme / Semester	B.E. Semester-I
Subject	Computer Programming using Python-I
Subject short name	Python-I
Test / CO	T1 / CO1
Offline / Online marks	16 / 9
Total T1 marks	25
Test date	Tuesday, 29 September 2026
Offline schedule	2:15 pm to 3:30 pm; scanning 3:30 pm to 3:45 pm
Online schedule	4:15 pm to 5:30 pm
QP deadline	Friday, 25 September 2026, 12:00 midnight
Submission email	qp.fyall@gmail.com

2. Branches and Subject Codes

Branches: CE / IT / CSD / AIML / AIDS / CSE / CST / CS&IT; / CEA / MA&CP; / CSE(AI) / CSE(DS) / CSE(CS) / CS&BIO / BIOTECH.

Subject codes: 017012194 / 017022194 / 017032194 / 017042194 / 017052194 / 017122194 / 017132194 / 017142194 / 017152194 / 017162194 / 017172194 / 017182194 / 017192194 / 017202194 / 017212194.

Mention all applicable subject codes in the official paper header.

3. T1 Syllabus and Faculty Blueprint

Unit	T1 topic	Full T1 blueprint (25)	Recommended offline target (16)
Unit 1	Introduction to Computer	2	2
Unit 2	Introduction to Python and Jupyter Notebooks	10	6
Unit 3	Conditional Execution and Iterations	13	8
Total		25	16

The faculty blueprint allows a unit-wise variation of 2–3 marks. Do not disclose the blueprint or chapter-wise allocation to students.

Out of T1 scope: functions, lists, tuples, dictionaries, sets, classes, files, modules, regular expressions, arrays, NumPy, Matplotlib, and other later-unit topics.

4. Mandatory Question-Setting Rules

- Question paper must follow the T1–T3 marks distribution issued by the HOD/FY authority.
- Ask conceptual and reasoning-based questions. Do not set easy recall questions.
- Do not ask definitions, formulae, theory short notes, True/False, Yes/No, or fill-in-the-blank questions.
- Difficulty level must be above average, while remaining solvable within the allotted duration.
- Q-1 must be compulsory MCQ.
- Maximum seven main questions are allowed; any number of sub-questions may be used.
- Mention all applicable branch subject codes.
- Do not use any LJJET faculty name in a question.

- Prepare three offline sets. Questions must be exactly the same in every set; only MCQ order may be reshuffled.
- Use only the approved official question-paper format. Do not redesign the template.
- Minimize pages while preserving clarity; normally use 11 or 12 point font as appropriate to the approved template.
- Do not save the QP on a personal PC, do not send it for prior verification, and do not print before completion of the examination.

5. Practice Book and Outside-PB Distribution

Component	Practice Book	Outside Practice Book	Total
Offline Python-I T1	10 marks	6 marks	16 marks
Online Python-I T1	0 marks	9 marks	9 marks

A PB question remains a PB question even if only its MCQ options are changed and the question stem remains unchanged. You may vary the marks of PB questions, but the total PB/outside-PB distribution must remain exact.

Faculty-only traceability requirement: Maintain a private mapping of every question to Unit, PB serial number or original source, marks, and final classification (PB / outside PB). Do not include this mapping in the student paper.

6. MCQ Rules

Rule	T1 requirement
Permitted MCQ marks	0.5 or 1
Number of options	Exactly 6 options for every MCQ
Mixed marks grouping	0.5-mark MCQs in Q-1(A); 1-mark MCQs in Q-1(B)
Output MCQ convention	Use e) Error and f) None of the above
Verification	Execute or manually trace every output-code question before finalising

Written-rule conflict requiring approval: The official instruction says that, when 0.5-mark and 1-mark MCQs are mixed, Q-1(A) and Q-1(B) should each have at least seven questions. The preferred pattern of 10×0.5 plus 5×1 conflicts with that instruction. Obtain written clarification from the HOD/FY QP authority before using five 1-mark MCQs.

7. Recommended Offline Paper Workflow

- Freeze the blueprint first: total 16; PB = 10; outside PB = 6; Unit 1 + Unit 2 + Unit 3 = 16.
- Select only T1 PB questions from Units 1–3 and record their serial numbers in a faculty-only audit sheet.
- Create outside-PB questions that test reasoning: operator precedence, type conversion, logical operators, nested decisions, while/for trace, break/continue, nested loops, validation, and counters.
- Check every MCQ for six options, one unique correct answer, code indentation, correct marks, and Bloom level.
- Prepare Sets A, B, and C with identical content; reshuffle only MCQ sequence.
- Create and verify an answer key for every set after reshuffling.
- Perform marks, syllabus, PB-source, and format audit before submission.

8. Practice Book Reference Examples

PB Sr. No.	Unit	Question type	Original marks
24	1	Algorithm and flowchart to calculate student percentage	3
26	1	Algorithm and flowchart for positive / negative number	3
33	1	Flowchart for Armstrong number	3
34	1	Flowchart for seconds to hour / minute / second	3
37	1	Flowchart for first N Fibonacci numbers	3
39–90	2	Python basics, expressions, variables, operators, type conversion, tracing	0.5 or 1
101 onward	3	Conditions, loops, nested conditions, break, continue, nested loops	0.5 or 1
195–201	3	Pattern programs	3 or 4

9. Online Paper Workflow

- Use one integrated 9-mark practical/system question under Q-3(A).
- The complete online question must be outside the Practice Book.
- Use Bloom level C / Create where applicable.
- Keep the problem strictly within T1 scope: variables, input/output, int(), arithmetic and comparison operators, if/elif/else, nested conditions, for, while, counters, validation, and break.
- Connect Unit 1 by requiring an algorithm or flowchart. Connect Unit 2 through variables, input, conversion, arithmetic, and output. Connect Unit 3 through conditions and iterations.
- Do not use functions, collections, classes, files, modules, or later-unit concepts.
- Place all practical-system requirements inside the Q-3(A) question cell; do not put faculty-only notes in the student paper.

10. Official Format Controls

- Use the original Python-I T1–T3 template file as the base document. Do not recreate its table structure from scratch.
- Preserve enrollment-number boxes, images/rectangles, merged cells, border design, row heights, column widths, tab stops, page breaks, and Bloom table.
- Populate only the existing template fields: Set, Test, CO, Offline/Online, branch, subject, subject codes, date, time, duration, marks, questions, marks column, and Bloom column.
- Use monospaced font and preserve indentation for all code blocks.
- Use monospaced font and preserve leading spaces for all patterns and required-output blocks.
- Use the department's bracketed marks format if that is used in the approved sample: [0.5], [1], [03], [09].
- Generate a PDF preview and visually check every page before calling any file final.

11. Faculty-Only Pre-Submission Audit

Category	Mandatory verification
Syllabus	Every question is restricted to Units 1–3 only.
Difficulty	Paper is above average; no recall-only or prohibited question type.
Offline marks	Total = 16 exactly.
Online marks	Total = 9 exactly.
PB split	Offline PB = 10 and outside-PB = 6 exactly.
Online source	Online question is entirely outside PB.
MCQs	Every T1 MCQ has 6 options and one verified answer.
Output MCQs	Options e) Error and f) None of the above appear where applicable.
Sets	Sets A/B/C contain identical questions; only MCQ order differs.
Header	Set, Test, CO, mode, branch, codes, date, time, duration, and marks are correct.
Format	Official template is used without structural changes; code and pattern indentation remain visible.
Confidentiality	No PB map, source serial number, chapter blueprint, answer key, or faculty-only note appears in student copies.
Submission	Send only approved student QP files to qp.fyall@gmail.com by the deadline.

Source basis: FY QP Setting Instructions T1–T3; FY Python-I faculty marks distribution; Python-I T1 Practice Book; Python-I T1–T3 official format; QP allocation document; Test-1 schedule. Faculty-only document.